
Name:

Entry No.:

Note: You may use facts that we have proved in the class without proving them again, but state such facts clearly and mention that you are using them because they were proved in the class.

1. [3.5 marks] Is the following statement correct? If yes, prove it. If not, argue why it is so with the help of an example.

If both ϕ and ψ are consistent, then $\phi \wedge \psi$ is consistent.

Recall that ϕ is said to be consistent if $\not\vdash \neg\phi$.

Space for rough work

2. **[3.5 marks]** Is the following statement correct? If yes, prove it. If not, argue why it is so with the help of an example.

If $\phi \wedge \psi$ is consistent, then both ϕ and ψ are consistent.

Space for rough work

3. [**3 marks**] Use propositional resolution to show the following.

$$(\neg p_1 \vee p_2), (\neg p_2 \vee p_3), (p_1 \vee \neg p_3), (p_1 \vee p_2 \vee p_3) \vdash (p_1 \wedge p_2 \wedge p_3)$$

Space for rough work

Space for rough work
